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DISTRIBUTION OF BLACK EARTHS IN BRAZILIAN AMAZON

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ABSTRACT

Patches of **Archaeological Dark Earth** (ADE) are found throughout the Amazon region. Although they are frequent, these soils do not appear on soil maps of the explorative or reconnaissance type, because they are usually of small dimensions. Most are less than 5 ha, but some occupy more than 300 ha. The thickness of the anthropic A horizon normally varies from 30 to 60 cm, but it may reach as much as 2 m in depth, with archaeological material (pottery, lithic artifacts, charcoal, etc.) present throughout the layer. ADEs are usually found on non-floodable terrains with well-drained soils. They are found in association with many soil types including latosols, argisols, ferralsols, podzols, ultisols, nitisols and petric plinthosols (Smith, 1980; Kern, 1988; Sombroek, Kern et al 2003; Falesi, 1970; Silva et al. 1970; Rodrigues et al. 1991, 2000). ADEs are associated with all of the archaeological phases proposed by Simões (1982), but they have higher representation in the Tropical Forest agriculture phase.

Key Words: Amazon soils; Anthropic horizon; Archaeology of Amazonia; Archaeological Black Earth, Terra Preta Arqueológica, Amazon archaeology.

INTRODUCTION

Soils known in the Amazon as Archaeological Dark Earth (ADE), Indian Black Earth, Black Earth, or Terra Preta contrast with the other soils of the region because of their high content of Ca, Mg, Zn, Mn, P and C. The organic matter of the ADE's is more stable and richer in Amazon organo-metallic components than the latosols (Pabst 1992). The high level of organic substance in ADEs can be attributed to the fact that they were formed during past human occupation of the site (Kern, 1996, Silveira et al 2002). In latosols the organic component is derived from the natural vegetal covering while in the ADE's the organic component is principally derived from the debris of human occupation including animal bones, turtle carapaces, shells, excrement, urine etc. in addition to vegetal components such as leaves for covering of houses, seeds, lianas etc. These components contribute to the formation of micro-ecosystems, with proper dynamics, that are the ADE's (Kern, 1996). These highly fertile micro-ecosystems formed in the past seem not to exhaust their chemical content in conditions of tropical forest. Therefore the ADE's are frequently sought by the local populations for the subsistence cultivation of cassava, banana, maize, papaya, etc., to obtain high productivity even without the use of fertilizer.

The high fertility of ADE soils, in addition to the stability of their nutrients, led to the development of several different kinds chemical and geochemical research including the definition of prehistoric settlement patterns (Kern 1988, Kern 1996 & Schann et al. 2001); study of the genesis of anthropic soils (Sombroek 1966, Kern & Kämpf 1989, Kern 1996); study of soil fertility (Pabst 1991; Woods & McCann 1999, Silveira et al 2002, McCann et al. 2000 & Glaser et. al., 2001), and the determination and behavior of the organic matter and the nutrient cycle in tropical soils (Sombroek 1995 & Ruivo et. al. 2001). Prehistoric artifacts found in ADEs, including ceramic fragments and lithics cause them to be registered as archaeological sites. In these sites the ADE's are used mainly to assess the size of the site and to determine the thickness of the occupation layer.

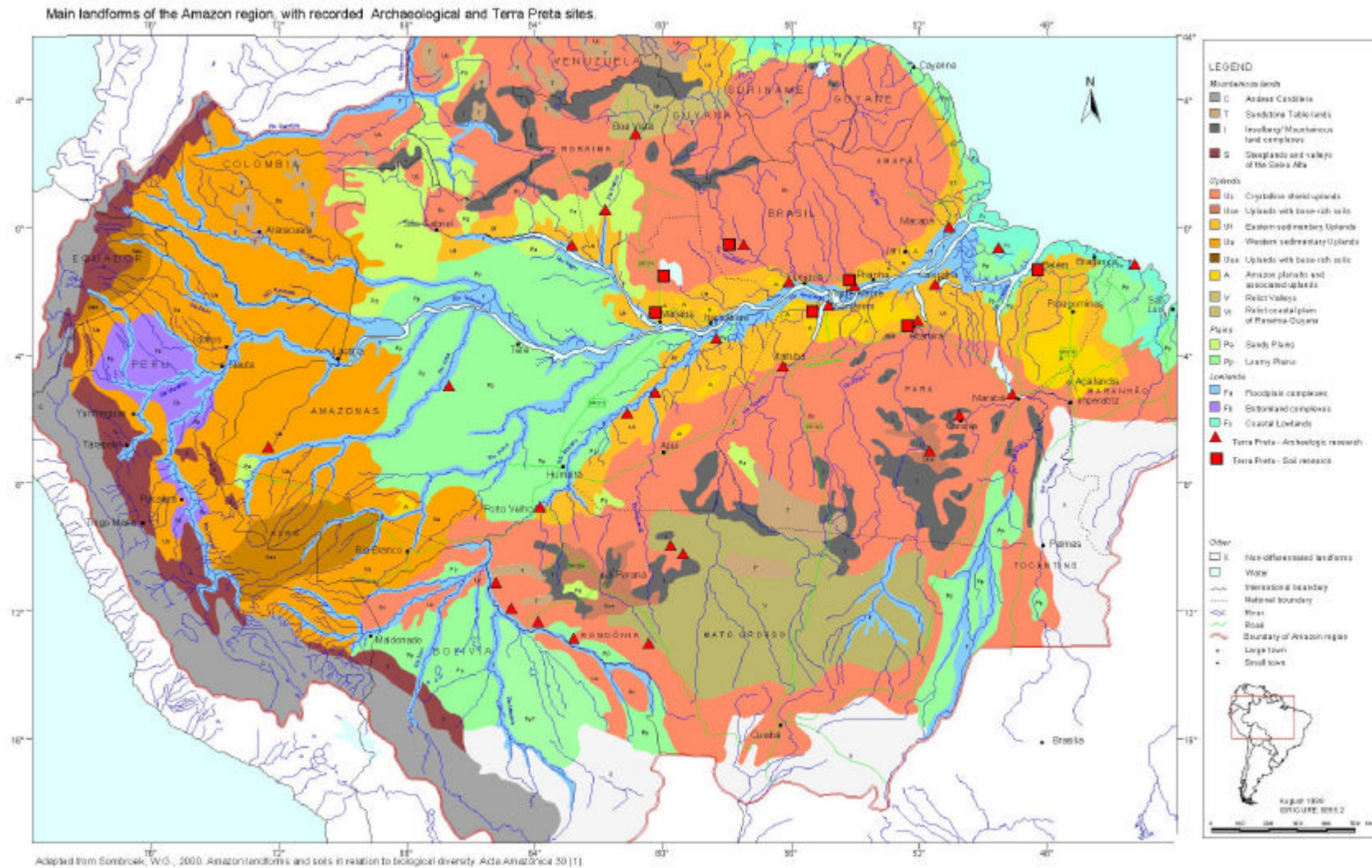
Because of the importance of the ADEs for many areas of the knowledge (archaeology, anthropology, pedology, geochemistry, mineralogy, etc.) this chapter will provide an overview of its geographic distribution, its positioning in the landscape, its relation with the existing soil classes in the Amazonia, as well as its relationship to prehistoric cultural periods.

LOCATION OF THE BLACK EARTHS (GEOGRAPHIC DISTRIBUTION)

The Brazilian Amazon is not a homogeneous region. It presents a remarkable variety of different aspects that allow it to be subdivided into 33 eco-regions in Brazil and 17 main landscapes in the Amazon region as a whole. Each of the eco-regions and landscapes has biotic and abiotic characteristics (Capobianco et al., 2001). Rivers, with their own peculiarities, constitute still different ecosystems. According to the water characteristics these rivers were classified by Sioli (1984) into three principal types: white, clear, and black waters. White water-rivers have their sources on the Andean slopes they have a great deal of material in suspension; are turbid, and are rich in fish. Native people use their fertile margins for seasonal crops. Clear water rivers run through areas with stable soils that originated from the crystalline rocks of the Guiana and Brazilian shields. The water is clear and transparent, having little material in suspension. Their nutrient content is in the middle between black and white water rivers. Rivers with black water are rich in humic acids, which produces the dark color of their waters. Meggers (1971) and Morán (1990) call them “starvation rivers” because they are not suitable for fishing, hunting, or food collecting on their margins.

The distribution of ADEs in the Amazon Region is enormous. They are present in practically all types of eco-regions and landscapes (Figure 1). A large number of archaeological sites are situated in the basins of white-water rivers such as the Madeira (Figure 2), Purus (Figure 3) and Uatumã (Figure 4) in Amazon State; Ji-Paraná (Figure 5) in Rondônia State; Tocantins (Figure 6) and Anajás-Marajó Island (figure 7) in Pará State; and along the Amazon River proper. There are also archaeological sites along clear water rivers such as the Trombetas (Figure 8), Mapuera/Trombetas (Figure 9), and Tapajós (Figure 10) in Pará State. Archaeological sites are even found along such black-water rivers as the Rio Negro (Figure 11) and Urubu (Figure 12) in Amazonas State; and Caxiuanã (Figure 13) in Pará State but the associated Terra Preta soil are mostly small in size. Smaller patches of ADE soils are also found in upper catchment areas, along the main tributaries of the Amazon, along rivulets (igarapés) feeding into these main tributaries such as former ring villages of the upper Xingu (Heckenberger et al. 1999), and along age-old forest tracts between river systems (McCann, 1999).

Recent geo-pedological surveys for ecologic-economic zoning in southeastern Amazonas and northeastern Rondonia yielded estimates of ADE sites every 5 km along the igarapés, and an over-all spatial coverage of one per 2 km² (Sombroek et al. 2002). Terra preta soils occur also on the lower Caquetá River in the Colombian Amazon (Andrade, 1986; Eden et al., 1984; Mora 1999), in southern Venezuela, and in Peru, but apparently less frequently.



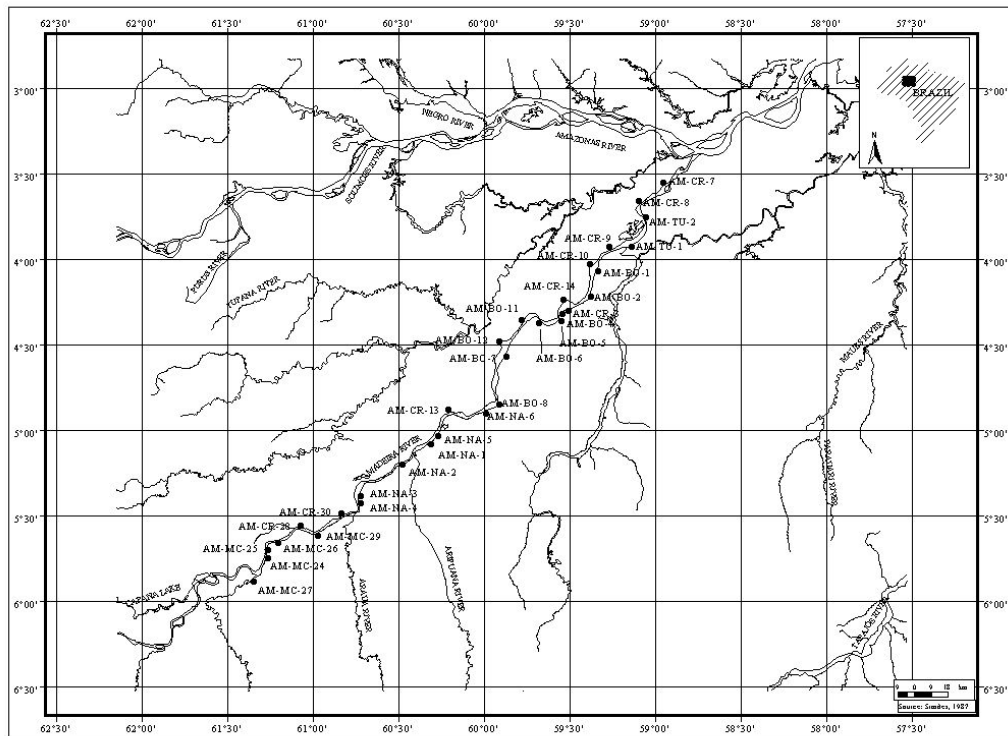


Figure 2 – Map of distribution of archaeological sites in the region of the Madeira river (Amazonas State) (Apud Simões & Lopes, 1987).

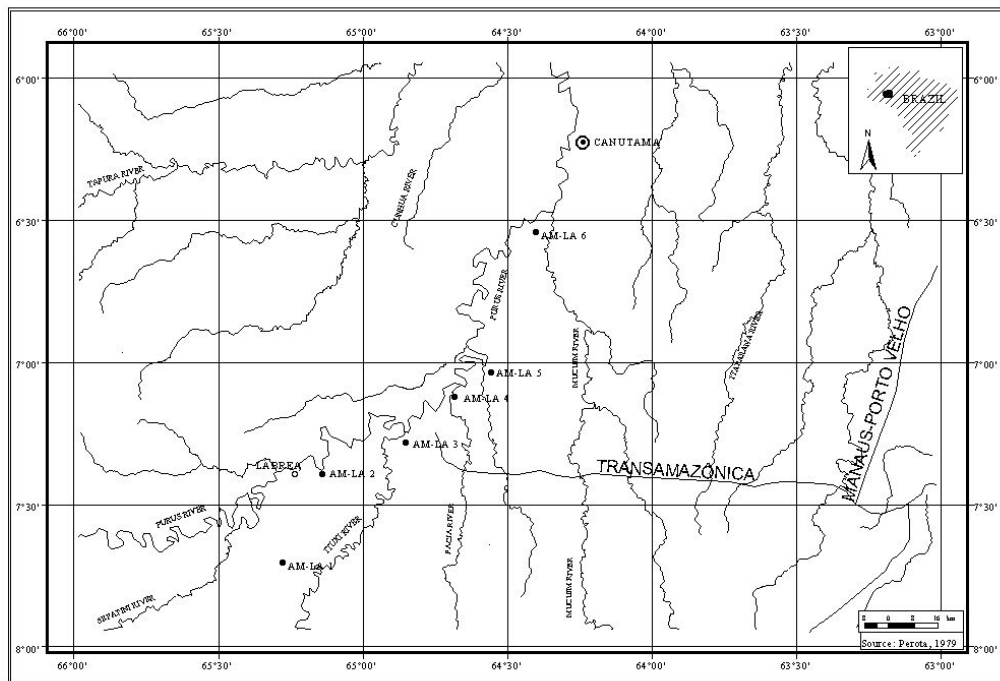


Figure 3 – Map of distribution of archaeological sites in the region of the Purus river (Amazonas State) (Apud Perota, 1979).

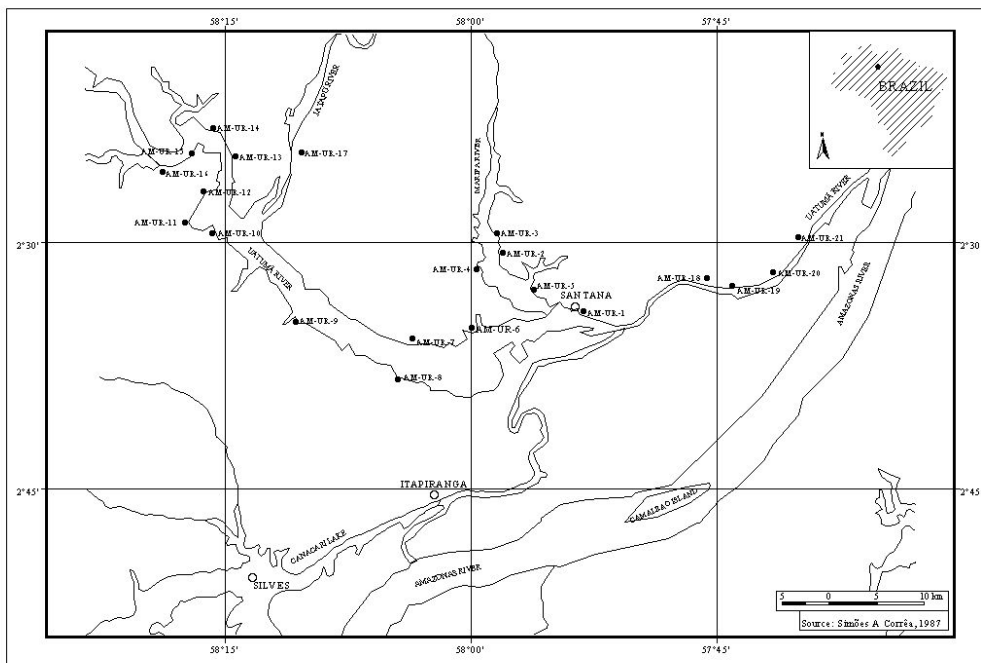


Figure 4 – Map of distribution of archaeological sites in the region of the Uatumã river (Amazonas State) (Apud Simões & Corrêa, 1987).

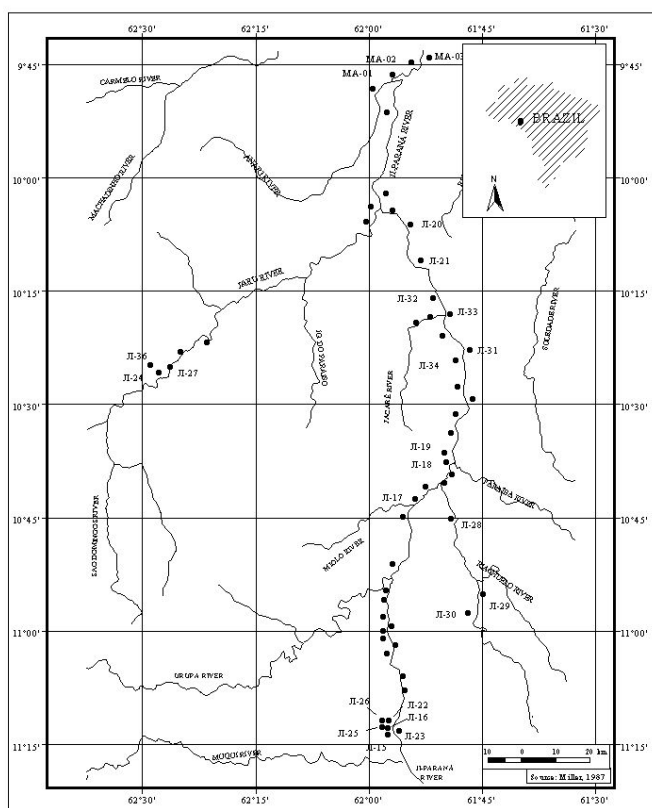


Figure 5 – Map of distribution of archaeological sites in the region of the Ji-Paraná river (Rondonia State) (Apud Miller & Caldarelli, 1987).

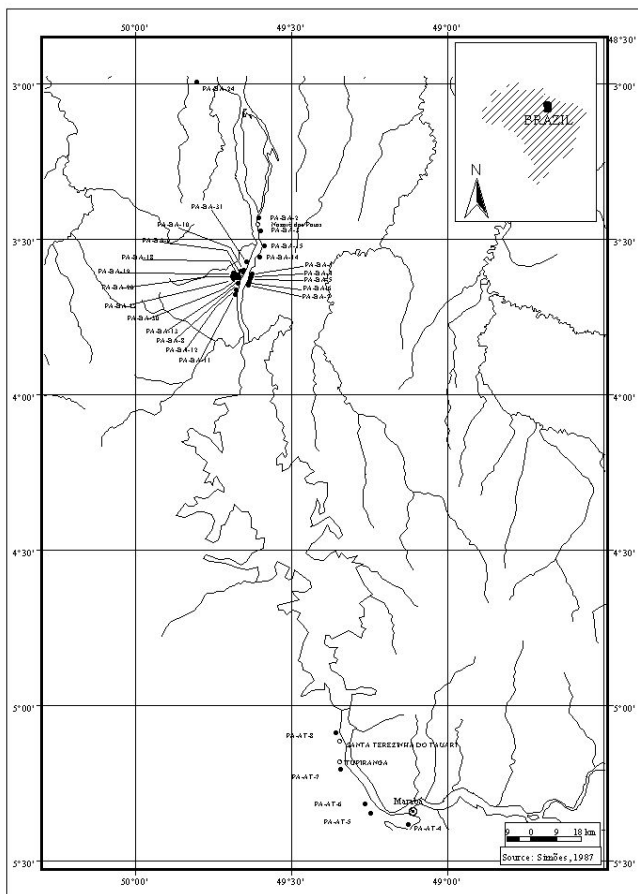


Figure 6 – Map of distribution of archaeological sites in the region of the Tocantins river (Pará State) (Apud Simões & Araujo- Costa, 1987).

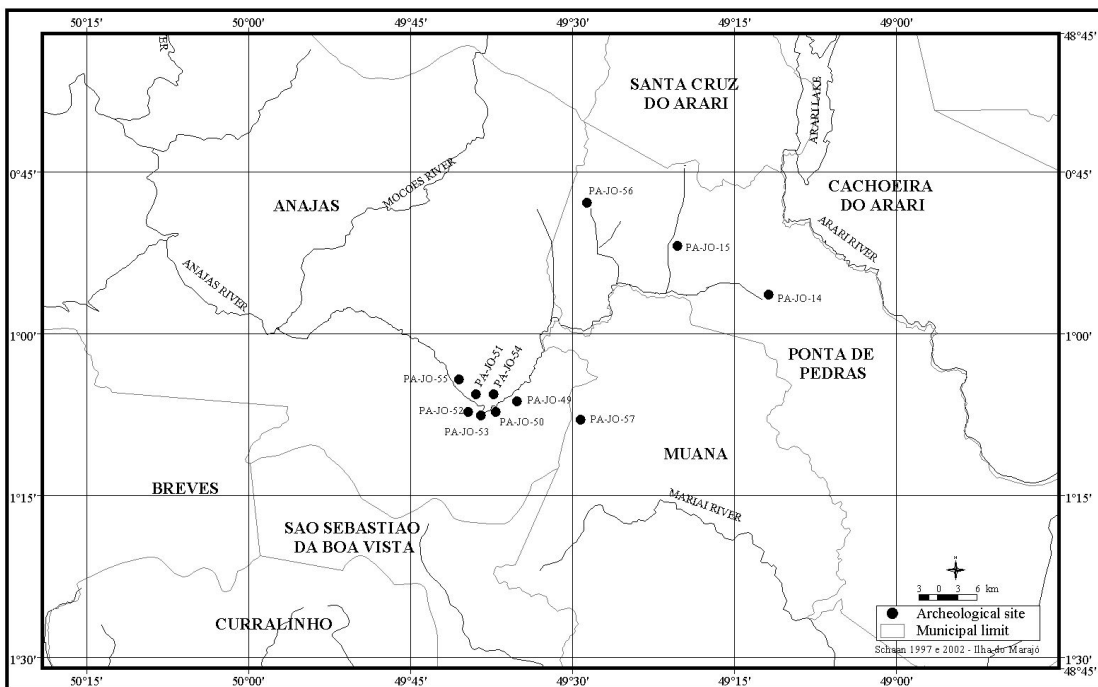


Figure 7 – Map of distribution of archaeological sites in the region of the Anajás river - Pará State (personal information of Denise Pahl Schaan).

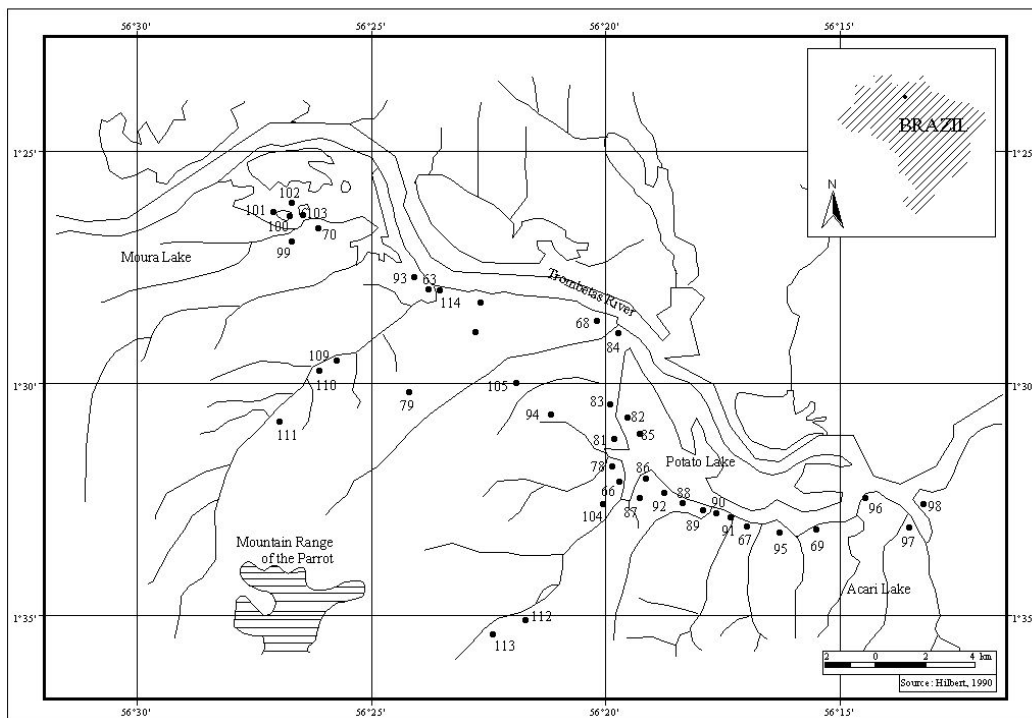


Figure 8– Map of distribution of archaeological sites in the region of the Trombetas river (Pará State) (Apud Hilbert, 1990).

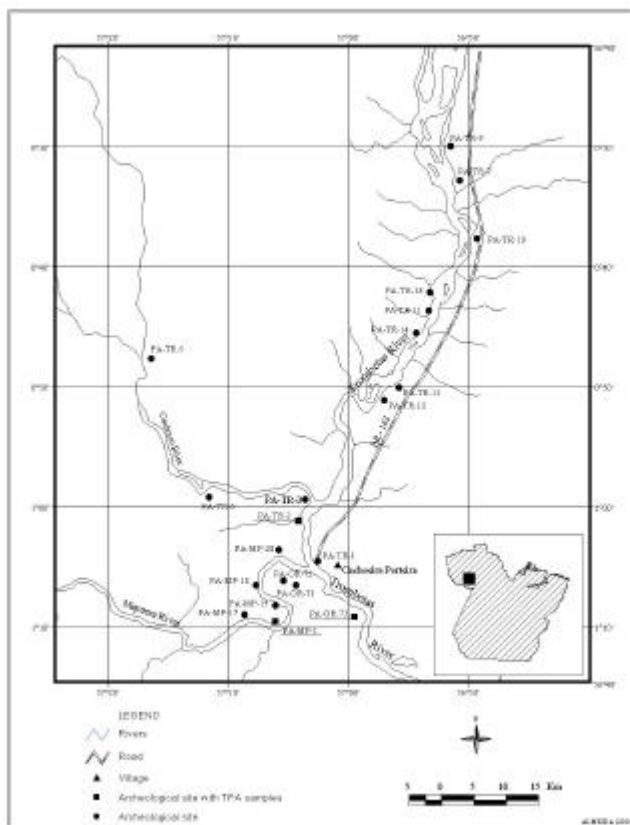


Figure 9 – Map of distribution of archaeological sites in the region of the Mapuera/Trombetas rivers (Pará State) (Apud Araujo-Costa, 1986 and Kern, 1988).

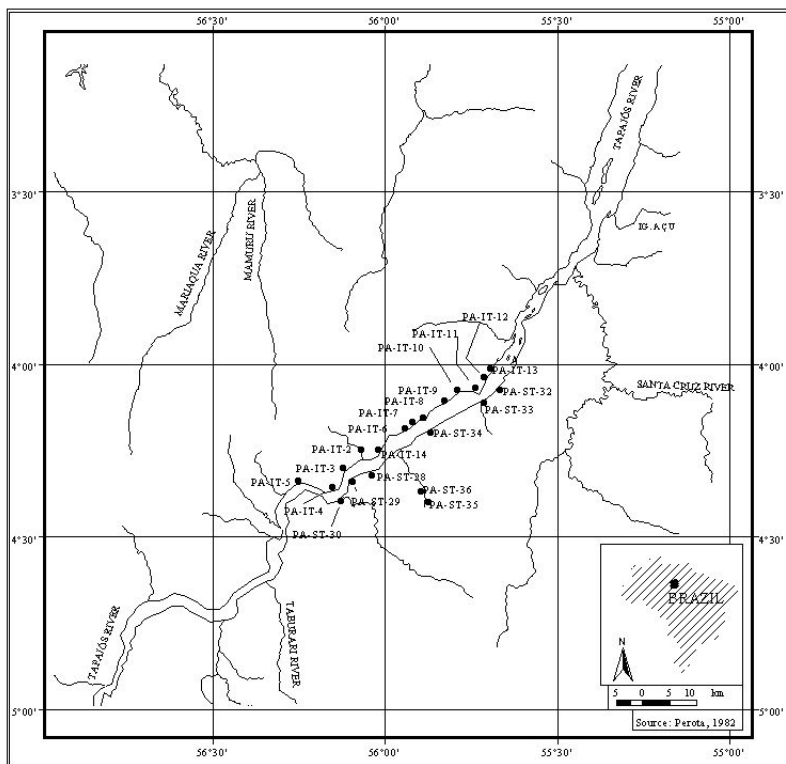


Figure 10 – Map of distribution of archaeological sites in the region of the middle Tapajós river (Pará State) (Apud Perota, 1982).

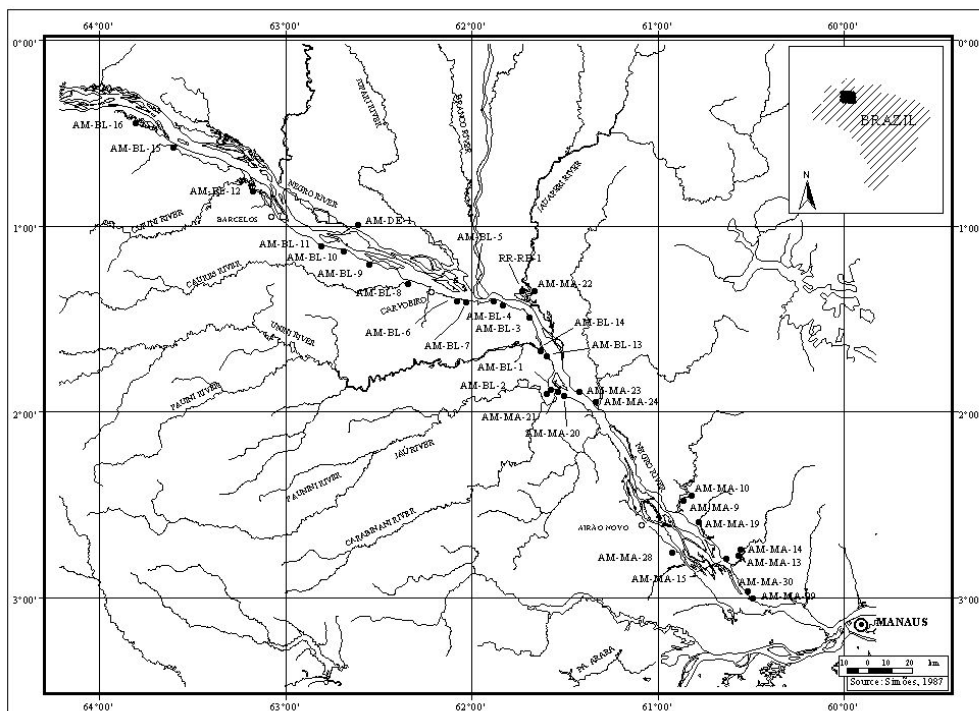


Figure 11 – Map of distribution of archaeological sites in the region of the Negro river (Amazonas State) (Apud Simões & Kalkmann, 1987).

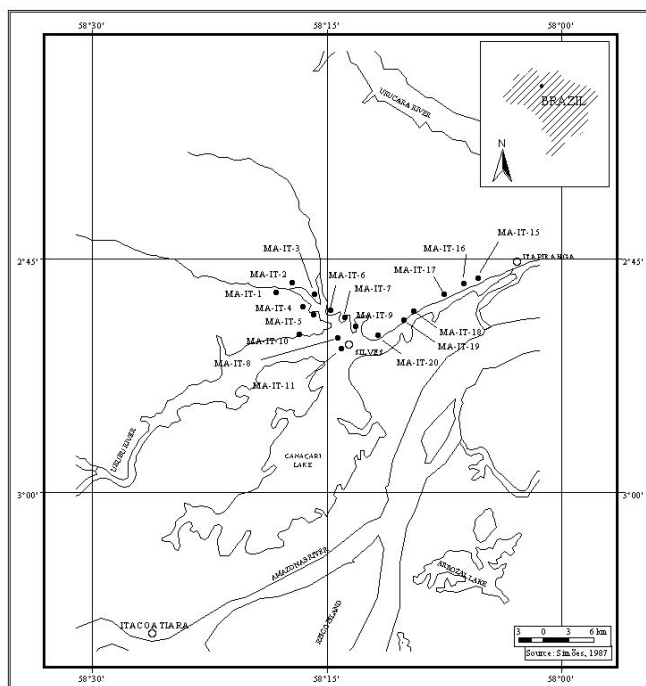


Figure 12 – Map of distribution of archaeological sites in the region of the Urubu river (Amazonas State) (Apud Simões & Machado, 1987).

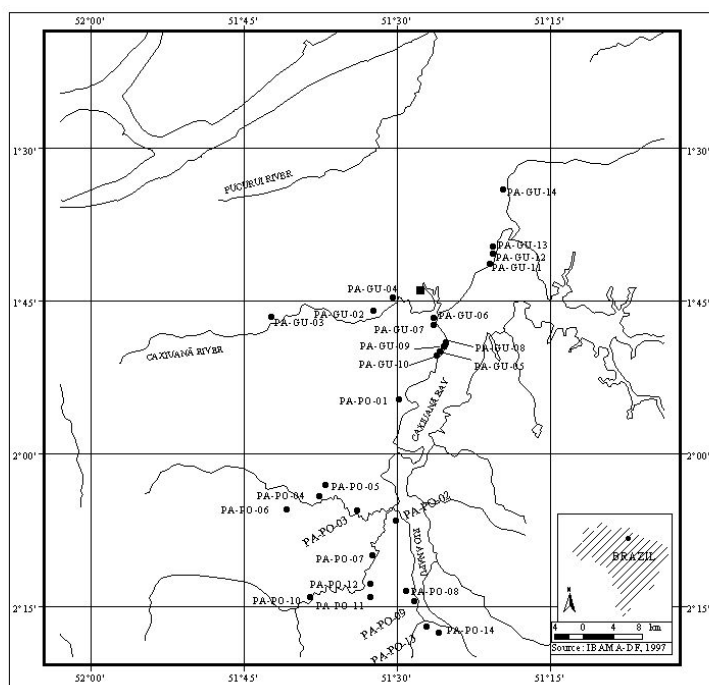


Figure 13 - Map of distribution of archaeological sites in the region of the Caxiuanã river (Pará State) (Apud Kern, 1996).

RELATIONSHIP OF THE “TERRAS PRETAS” WITH THE LANDSCAPE

Archaeological papers reveal that most sites are located near the confluence of streams and rivers, or near falls or rapids but the lack of access roads makes interior locations extremely difficult to evaluate.

Hartt (1885) was the first to report the existence of black earth on the plateau south of Santarém. Nimuendaju plotted these sites on his map and recorded the existence of straight roads 1.5m wide and by 30 cm high that linked dark earth sites with each other. He also noticed the presence of ponds built by ancient populations, perhaps because of the long distance between the rivers and the plateau and the absence of rivulets on the plateau itself (1949; Palmatary 1960: 20, map 4). During 1985-1986 archaeological projects on Xingu and Trombetas rivers developed a systematic research program using semicircular sample units with a radius of 5 km in conjunction with the classic methodology of identifying sites from information provided by local informants. In spite of the sophisticated survey approach, the project found most of the archaeological sites were situated on the riverbanks and were identified thanks to the information of local people (Barreto, 2001: 69-71; Kalkmann, et al. 1985a, 1985b).

These investigations demonstrated that 45% of the archaeological sites were situated between 5 and 25 m above the water sources (Figure 14). Only 4% of the sites were located on areas over 40 m high including sites in Oriximiná and Santarém regions of Pará State. These sites were on the edges of plateaus, places of difficult access, far from water sources, that may have been occupied because of demographic pressure. Archaeological sites with ADE were also found in areas that are seasonally flooded but only 1% of the sites are located less than 2 m above nearest water sources. Such sites may have been seasonal use sites, perhaps the fishing grounds of prehistoric people (Nimuendaju 1949).

Elsewhere there were some large communities were in the floodplain. The Orsua expedition found an Omagua village of thirty multi-family dwellings on an island near the mouth of the Napo River in 1560. By 1698 all of the Omagua settlements were on islands for protection from hinterland peoples such as the Ticuna (Myers 1992b). Peoples of the central Amazon also lived in island communities at the end of the seventeenth century. On Careiro island, at the mouth of the Negro River, there are sites with non-calibrated radiocarbon dates of 100 B.C. and A.D. 850 (Sternberg 1960: 419). In February 1561 the remnants of the Orsua expedition stopped at a Tapajós island village, possibly a fishing community, shortly before reaching the principle towns on the high ground at the mouth of the Tapajós River (Ortiguera, 1981: 111). A dark earth site on Ituqui Island, 30 km below Santarém (WinklerPrins, 1999), also may be referable to the Tapajós. On the várzea of the Nhamundá River a *terra preta* site of the Barrancoid [Incised Rim] Tradition covers an area of some 4 ha. Three calibrated radiocarbon dates suggest that the site was occupied between A.D. 7 and A.D. 247 (Hilbert and Hilbert, 1980: 2; Simões and Araujo-Costa, 1978: 109).

The only systematic investigation for archaeological sites in the interior took place along the segment of the Transamazon highway between Altamira on the Xingu River and Itaituba on the Tapajós River (Smith 1980). Half of the sites on this transect were more than 200 m from a perennial stream or river (table 1*). They ranged from 0.5 to 5.0 ha in size. Two sites on the banks of a perennial stream ranged from 0.3 to 0.7 ha, but all of the sites on a major river, the Xingu or Tapajós, exceeded 10 ha. This tiny sample suggests that more than half of the ADE sites are located far from a major river. Sites on a major river also seem to be far larger than sites near minor streams in the interior or hinterland although there is certainly some overlap.

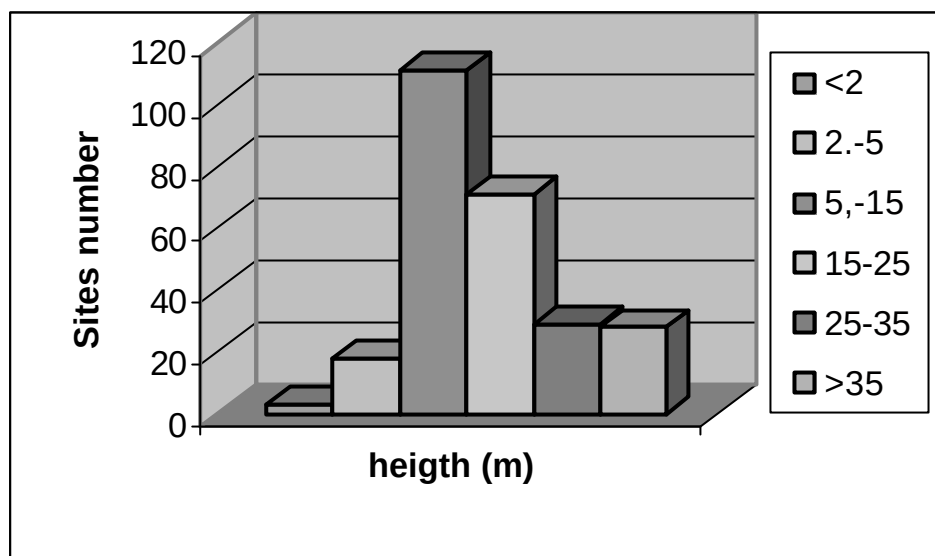


Figure 14 - Height of ADE Archaeological sites in relation to the nearest water sources.

These results of the Xingu and Trompetas surveys confirm that the ADEs are generally located on non-floodable land (“terra firme”). They are on well drained soils located next to running water, and they were usually in a topographic position that allowed for a wide view of their neighborhood, which was an strategic and important factor for survival of prehistoric man. These places have sometimes been re-used by contemporary people for habitation or for agricultural purposes. For example, the modern city of Belém do Pará was founded on a little hill on the margin of Pará River in 1616. First there was a fort, initially called “Presépio” and later “Forte do Castelo”, established to preserve the village from possible invaders. Recently a group of archaeologists recovered the remains of the ancient occupations and observed that under a layer of 70 cm of modern garbage was a layer of ancient terra preta. It is very common to find deposition of different occupations on strategic places because ancient people as well as contemporary ones look for some specific characteristics in the landscape, that allow to build fortresses, harbors and agricultural occupation. In smaller present-day villages, ADE patches are often found and by implication maintained under primitive football fields in the centre of built-up areas.

RELATIONSHIP OF THE “TERRAS PRETAS” WITH OTHER SOILS

ADEs are found on a variety of different soil types including latosols, argisols, espodosols, plinthosols, nitosols, cambisols, and orthic quartzarenic nitosols (Embrapa, 1999), as may be observed on the map (Figure 1) and from Table 1 (Embrapa, 1981, IBGE/SUDAM, 1990). ADEs on latosols and agrisols are the most common. They extend over larger areas and are located on lands that do not flood seasonally. They are well drained and very thick. Their texture varies from sandy to heavy clayey, with low reserves of nutrients that are essential for crops (Silva et al, 1970; Kern, 1988; Rodrigues et al, 1974; 1972; 2000; 2001 and 2002; Rodrigues, 1996). ADE soils are less frequent on orthic quartzarenic nitosols, ferrocronic espodosols, red nitosols, and cambisols.

The anthropogenic transformation of the superficial horizons of dystrophic soils with base saturation less 50 percent because of development of “Terras Pretas” produced the eutrophia the these horizons with base saturation more 50 percent. Sometimes that process occurs in the whole profile because of the Ca movement which increases its concentration, diminishing or eliminating aluminium from the cation exchange

sites. Analysis of ADE from Santarém, Caxiunã, Itaituba, Monte Alegre, Oriximiná and Marajó in the Pará state, and from Iranduba, Manacapurú, and Apuí in the Amazonas state (Table 2), demonstrates that the proportion of Ca, P, K, Mg, and the organics C and N, is relatively high in the upper layers of ADE and generally decreases with depth. The nutrient content of ADE sites is quite variable with contents of exchangeable Ca, Mg and K, ranging between 0.5 to 48.48 cmolc kg⁻¹ of soil and 0.10 to 16.46 cmolc kg⁻¹ of soil, and 0.01 to 0.66 cmolc kg⁻¹ of soil, respectively. P contents, determined by Mehlich method (Embrapa, 1997), varies from 42 to 7,455 mg kg⁻¹ of soil and organic carbon; and nitrogen contents between 1.40 to 55.60 g kg⁻¹ of soil and 0.20 to 4.90 g kg⁻¹ of soil, respectively. The pH micrease with the formation of ADE and varies between 4.6 to 6.6.

The textural classes of these soils vary from sandy to heavy clayey with contents of sandy, silt and clay fractions between 125 to 840 g kg⁻¹ of soil, and 50 to 480 g kg⁻¹ of soil and 50 to 780 g kg⁻¹ of soil respectively (Table 2).

Even though ADEs are spread throughout the region, because of their small size they are normally plotted as inclusions on the soil maps created as results of exploratory survey (Kern, 1988). Eighty percent of the (individual) ADE's are less than 2 ha in size (Figure 15) though a few cover more than 100 ha.

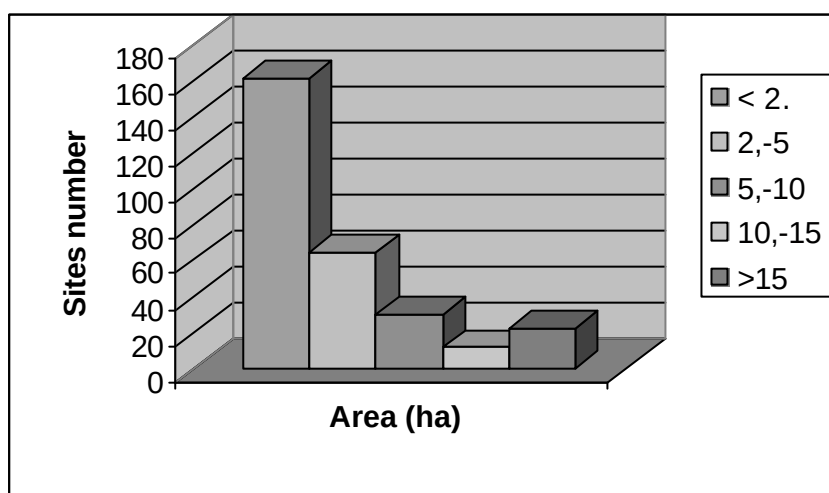


Figure 15 – Size of archaeological sites.

Fifty-seven percent of the registered archaeological sites with anthropogenic A horizons are from 30 to 60 cm in depth (Figure 16) , well structured and dark brown to black in color (N2/ 5YR 2/1, 7.5YR 3/1 to 10YR 3/1). They are significantly deeper than the forest soils with A horizons, which are only 10 to 15 cm in thickness. The black brown to black color of the anthropogenic A horizon (Table 2) it is the result of the melanization process caused by the addition of organic material in the superficial horizon by formation of organic metallic (complexes) compounds mainly Ca-humate.

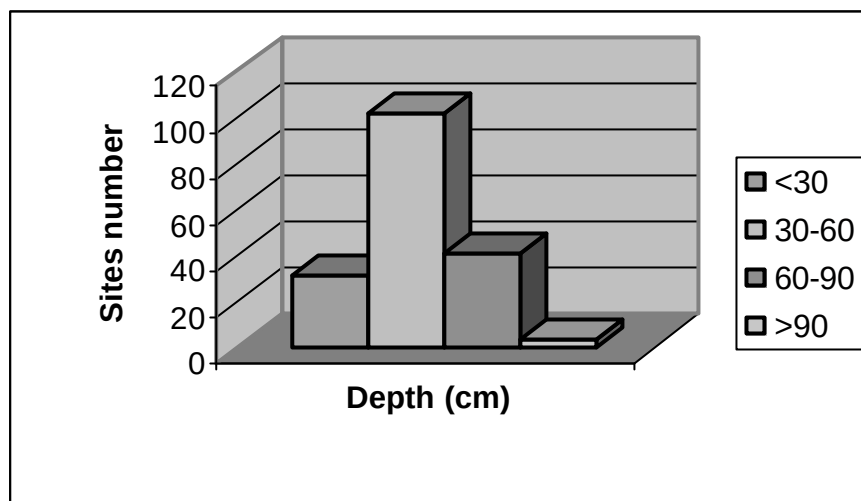


Figure 16 – Depth of A -horizon in archaeological sites of ADE

The morphological, physical, and chemical characteristics of anthropogenic soils, as well as their depth, may differ within a single site or among diverse sites. These variations may be directly related to the prehistoric community pattern. For example, the Manduquinha site in Caxiuanã Region of Pará State was occupied for over 300 hundred years, from A.D. 1280 to 1600. The characteristics of ceramic manufacture, temper, firing, and decorative techniques suggest cultural continuity throughout this period although the occupation of the site may not have been continuous. The small extent of the ADE (0.5ha), the limited depth of the occupational layer, and the limited quantity of archaeological material suggests that only a small group was present (Kern, 1996). In contrast, the great extent of archaeological sites in the upper Xingu region indicates that greater demographic density existed in that region (Heckenberger 1998).

TRANSITIONAL SOILS: “TERRAS MULATAS”

In a number of cases the patches of ADE are surrounded by a broad band of Terra Mulata (TM) soils. These soils are dark greyish brown rather than black in their upper horizons, have less elevated P levels, and no or very few artifacts. Still they have elevated levels of soil organic matter in comparison to undisturbed forest soils. TM soils have been described by Sombroek (1966) for the Arapiuns area west of Santarém, for the Santarém-Belterra area by McCann (1999), by Horbe (verbal comm.) along the Rio Juma, 100 km north of Apui on the Transamazonica, by Jarbas Cunha (in prep.) for the Humaitá area on the Madeira river, and by Sombroek (in prep.) for terrace lands along the Jamanxim river near Novo Progresso. These “transitional” soils are also known for the Colombian Amazon where they are called *tierras pardas* (Mora, pers.). These authors consider such TM areas to be pre-Columbian agricultural fields that surrounded former major Indian villages. The characteristics of these soils are intermediate between the kitchen middens of ADE sites proper, and the unaltered soils of the forest where only foraging took place. Whereas the increase in soil organic matter content and mineral fertility on the homesteads may have taken place accidentally, the increase in TM areas would be the result of the intentional application of human or animal manure, green manure from pounded mollusc shells, or root accretions. This application would allow a semi-permanent horticultural type of food production once a piece of forest-land had been cleared initially with stone hand-axes and subsequent burning.

Thus far, very few TM soils have been systematically described and analysed, hence their distinction in quantitative terms from the ADE soils, *sensu strictu*, on the one hand, and from the natural soils on the other hand, is still to be established. A provisional map of their occurrence in the Santarém – Belterra area, in association with known ADE (figure 17).

Both ADE and TM soils are surrounded by natural soils formed under primary forest, where most of the organic matter is concentrated in the upper 20 cm. They may be ferralsols (mainly xanthic, sometimes orthic or rhodic), acrisols or else, have a sandy to clayey texture, variable stoniness, and almost invariably a

high amount of exchangeable Al (with pH 4.5 or lower). A technical reference collection of whole soil profiles of ADE, TM, and their non enriched relatives is being established at the Coordenação de Ciências da Terra e Ecologia of MPEG, Belém, in cooperation with the local Empraba center CPATU.

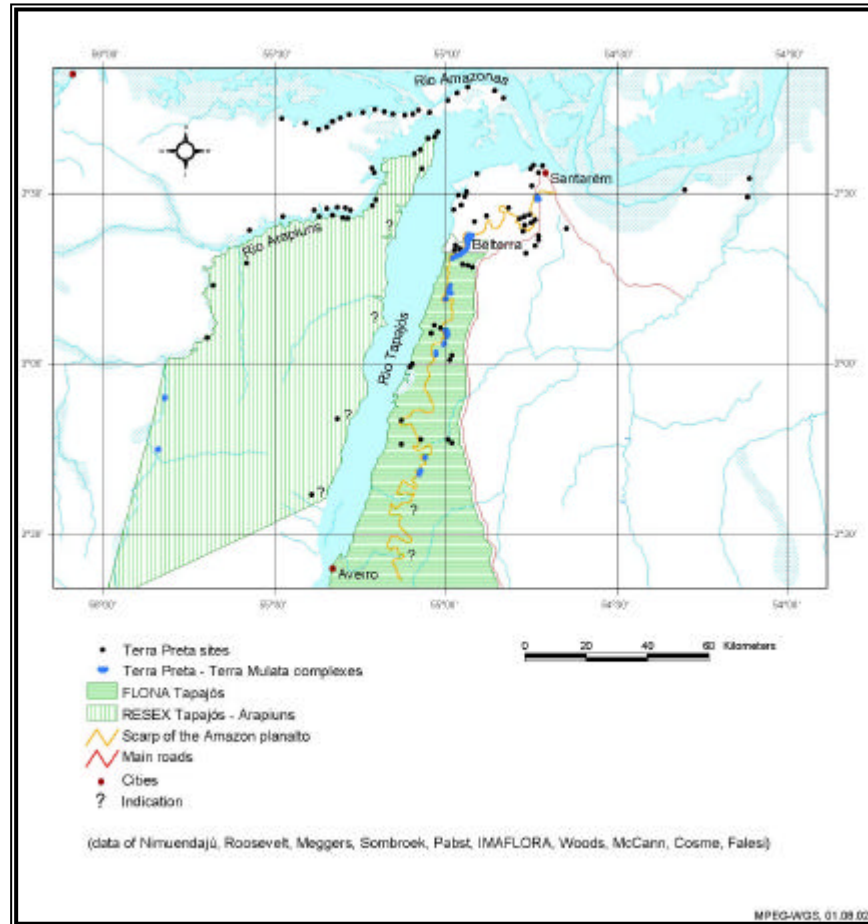


Figure 17 - Occurrence of TM in the Santarém – Belterra area, in association with ADE (Sombroek et al 2002)

RELATIONSHIP OF THE DISTRIBUTION OF TERRAS PRETAS WITH CULTURAL TRADITIONS

During mid twentieth century Betty Meggers developed an explicative model of Amazonian cultural development, (Meggers, 1954; Meggers & Evans, 1957; Meggers, 1971) based upon Steward's model of cultural ecology (1963) and White's model of cultural materialism (1945). Meggers argued that the environment was determinant for cultural development and for food production. She proposed an environmental classification based on the agriculture potential and cultural diversity, establishing four different types: first, lands without agricultural potential; second, areas of limited agricultural potential; third, lands with considerable agricultural potential, and fourth, lands having an unlimited potential. Meggers linked agricultural potential with Steward's (1963) cultural classification into Marginal, Tropical Forest, Circum-Caribbean, and Andean areas. According this classification the tropical forest has limited agricultural potential because the soils are poor due to their acidity, because they are derived from sedimentary and ancient rocks that are poor in nutrients, and because of the high temperatures and high rainfall that are characteristic of the tropical forest. The poverty of environmental sources would limit indigenous culture

development, stop population growth, economic intensification, and ethnographic specialization (Meggers, 1954; Meggers & Evans, 1957; Meggers, 1971). According to this anthropological vision the Amazon environment would not permit the development of natural resources on a scale sufficient to permit subsistence of dense populations and permanent settlements.

Within this theoretical framework, the complexity of Marajoara culture presented a special problem. The size of the artificial mounds of Marajó, the complexity of the pottery, and the unusual settlement pattern all suggested a social organization typical of the Circum-Caribbean region rather than of the Amazon. Meggers suggested that Marajoara must have originated in northwestern South America, outside of the Amazon region. She believed that it reached Marajó at the peak of its development where it inevitably decayed because of the debilitating environment conditions of the tropical forest (Meggers and Evans 1957: 593-4).

Meggers summarized the wide range of ceramic variation among Amazonian cultures by postulating four ceramic horizons (Zoned-Hachure; Incised Rim; Polychrome; and Incised and Punctate), stylistic groupings that are still widely accepted by most Amazonian archaeologists. She suggested that the Zoned-Hachure, Polychrome, and Incised and Punctate horizons had originated in the Andes and immigrated into the Amazon Basin where they inevitably declined. She is unclear about the origins of the Incised Rim Horizon, noting only that sites were distributed from the Amazon to the Orinoco (Meggers & Evans, 1961).

In contrast, Lathrap (1975) believed that the ceramic horizons identified by Meggers and Evans were better understood as traditions with considerable time depth rather than as horizons with considerable geographical spread but little time depth. He also parted company from Meggers in believing that rich várzea of the central Amazon had spawned the complex cultures characterized by the Zoned-Hachure, Incised Rim [Barrancoid], and Polychrome ceramic traditions. Population growth in the central Amazon had led these cultures to radiate outwards, upriver to Bolivia, Perú, and Colombia. Earlier inhabitants had been pushed to the margins of the Amazon by later arrivals, but some of the descendants of the early migrants still existed at least until the European invasion.

Simões' (1982) reconstruction of Amazon prehistory suggests that the region was first occupied by hunter-gatherer preceramic groups (10,000 – 1000 BC), followed by fisher-gatherer-ceramic groups (3200-200 BC), then by tropical forest agriculturalists (0- A.D. 1600), and finally by Sub-Andean Agriculturalists (A.D. 100-1300). These four cultural phases, encompassed all the styles horizons proposed by Meggers & Evans (1961) and seconded by Lathrap (1975).

Roosevelt (1991) proposed a new cultural sequence for the Amazon based on ethnographic, archaeological, anthropological, and paleontological evidence. Like Lathrap, she disagreed with Meggers about the environmental limitations of the Amazon and suggested that long term occupations were common and that complex societies developed locally. She suggested (1991) the Amazon was first inhabited by Paleo-Indian groups (11,200 BP) whose subsistence was based on hunting, fishing, and gathering. They were followed by hunting - fishing and mollusk gathering groups that established themselves along the Amazon River (7500-4000 BP). These groups knew the pottery manufacture, though in a rustic form. Afterwards came the roots – horticulture societies (4000-3000 BP), which manufactured a more elaborate pottery. At the end of the sequence were the Marajó mound - builders (1000 BC) who developed into complex societies linked to the Polychrome Horizon and Incised-and-Punctate ceramic styles.

Permanent settlements and complex societies as proposed by Lathrap (1975) and Roosevelt (1991) are endorsed by other archaeologists including Brochado (1970), Myers (1992a), and Heckenberger (1998). Most are associated with ADEs. The dimensions of occupational refuse in ADEs (depths and extension of inhabited areas) are directly related to the length of settlement and the density of population. ADEs are present in all Amazon cultural horizons established by Meggers & Evans (1961) from the oldest Zoned-Hachured Tradition (980 ? 200BC) up to the recent Incised and Punctate Tradition (A.D. 480 up to European contact). They occur through all the cultural phases proposed by Simões (1982), with the exception of the pre-ceramic hunters-gatherers, but the greatest representation of these soils is present during the period of tropical forest agriculturalists (0 - A.D. 1600). This phase includes the Santarém Culture on the Tapajós River; the Konduri Complex on the Nhamundá and Trombetas Rivers; the Mazagão Phase in Amapá; and the Paredão Phase on Rio Negro in addition to other cultural traditions in Xingu and Tocantins rivers.

Conclusion

- ADEs have a broad distribution in Amazonía, being present in all eco-regions and landscapes.
- Besides the Brazilian Amazon there are occurrences in other Latin-American countries including Colombia, Perú, Venezuela, and the Guianas (Andrade , 1986 and Eden et al.). Apparently prehistoric man who built the ADE was able to have their settlements all over the different Amazon environments.
- ADEs are usually located on the well-drained soils of “terra firme,” near to water sources, and always on a topographic position able to have a broad vision of their neighborhood. The landscape played an important role on the decision to install their settlements, with security for the prehistoric groups survival – as a key element.
- “Terras Pretas” and the associated “Terras Mulatas” are found on the most different types of soils: from the strongly weathered Latossolos, considered, leached up to the Red Nitosols which present high natural fertility. So the soil must not have been the determinant factor for the establishment of prehistoric groups.
- ADE soils vary with respect to the depth and extension of the occupation area as well as their physical and chemical characteristics among sites and even within a single site. The quantity and types of the settlement refuse in ADE are related to the length of the occupation and density of the population living on it.
- Black Earths are present in all of the cultural phases proposed by Simões (1982), but are particularly well represented on sites of the tropical forest agriculturalists.

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Table 1*. Location and characteristics of ADE sites on Altamira – Itaituba [Xingu R to Tapajós R] transect (after Smith 1980: table 2).

Site #	Size of site 200m+ from waterway	Size of site adjacent to river or stream	Depth of ADE	Distance from nearest waterway	Size and shape of ADE site	Artifact distribution
1	1.5 ha		0.40	300m from perennial stream, Xingu-Tapajos	Roughly circular	Sherds abundant to 15cm
2	5.0		0.20	300m from perennial stream, Xingu-Tapajos	Roughly circular	Sherds at 10-20cm; stone ax
3		10.0 ha	0.20	Rt bank, Tapajos	1km along bank, 100m inland	Sherds and axes numerous to 20cm
5	1.0		0.40	400m from perennial stream, Xingu-Tapajos	Roughly circular	Sherds to 10cm
6	0.5		0.87	200m from perennial stream, Xingu-Tapajos	Roughly circular	Sherds at 15cm
16		0.3	0.15	Perennial stream 30m away	Lenticular	Sherds abundant to 15cm
17		90.0	0.68	Xingu R	1.8km along bank and 500m inland	Sherds and celts abundant to 10cm
27		15.0	0.45	Itaituba, Tapajos R	1km along bank, 150m inland	Sherds rare from surface to 15cm
28		0.7	0.35	Margin of perennial stream	Lenticular	Sherds numerous from surface to 10cm
29	1.6		0.25	Perennial stream 200m away, Xingu-Tapajos	Lenticular	Sherds numerous to 15cm

Table 1. Soil classes distributions on sites of the anthropogenic ADE

STATE	SITES	SOIL CLASSIFICATIONS Brazilian System (Embrapa, 1999) adapted Brazil soil map (Embrapa 1981) Amazonia soil map (IBGE/SUDAM, 1990)
Amazonas	Ipixuna	ARGISSOLO AMARELO Distrófico típico, textura média/argilosa + ARGISSOLO VERMELHO Eutrófico típico, textura média/argilosa.
	Carauai	ARGISSOLO AMARELO Distrófico típico, textura média/argilosa + ARGISSOLO AMARELO Distrófico plântico, textura média/argilosa + LATOSSOLO AMARELO Distrófico típico, textura argilosa.
	Bacelos	ARGISSOLO AMARELO Distrófico típico, textura média/argilosa + PLINTOSSOLO HÁPLICO Distrófico típico, textura argilosa + ESPODOSSOLO FERROCÁRBICO Hidromórfico típico, textura arenosa.
	Presidente Figueiredo	LATOSSOLO AMARELO Distrófico, textura argilosa + ARGISSOLO AMARELO Distrófico típico, textura média/argilosa
	Itacoatiara	LATOSSOLO AMARELO Distrófico típico, textura muito argilosa + ARGISSOLO AMARELO Distrófico típico, textura média/argilosa.
	Irlanduba	ARGISSOLO AMARELO Distrófico típico, textura média/argilosa
	Manacapuru	LATOSSOLO AMARELO Distrófico + ESPODOSSOLO FERROCÁRBICO típico, textura muito argilosa + ESPODOSSOLO FERROCÁRBICO Hidromórfico, típico, textura arenosa
	Manicoré	LATOSSOLO AMARELO Distrófico típico, textura média + NEOSSOLO QUARTZARÊNICO + LATOSSOLO VERMELHO AMARELO Distrófico típico, textura argilosa + ARGISSOLO VERMELHO AMARELO Distrófico típico, textura média/argilosa.
	Apuí	LATOSSOLO AMARELO Distrófico típico, textura argilosa e muito argilosa.
	Labrea	ARGISSOLO VERMELHO AMARELO Distrófico típico, textura média/argilosa + ARGISSOLO AMARELO Distrófico plântico, textura média/argilosa + ESPODOSSOLO FERROCÁRBICO Hidromórfico típico, textura arenosa
	Novo Aripuanã	LATOSSOLO AMARELO Distrófico típico, textura média + NEOSSOLO QUARTZARÊNICO Órtico típico.
	Rio Roosevelt	LATOSSOLO AMARELO Distrófico típico, textura argilosa + ARGISSOLO VERMELHO AMARELO Distrófico típico, textura média/argilosa.
Rondônia	Porto Velho	LATOSSOLO AMARELO Distrófico típico, textura argilosa.
	Jaciparaná	LATOSSOLO VERMELHO AMARELO Distrófico típico, textura argilosa.
	Guajará Mirim	NEOSSOLO QUARTZARÊNICO Órtico, típico + NEOSSOLO LITOLICO Distrófico.
	Pacaas Novas	LATOSSOLO AMARELO Distrófico típico, textura média + ARGISSOLO VERMELHO AMARELO Distrófico típico, textura média/argilosa + NEOSSOLO QUARTZARÊNICO Órtico típico
	Príncipe da Beira	LATOSSOLO AMARELO Distrófico típico, textura média + NEOSSOLO QUARTZARÊNICO Órtico típico + ARGISSOLO VERMELHO AMARELO Eutrófico típico, textura média/argilosa.
	Pedras Negras	LATOSSOLO AMARELO Eutrófico típico, textura média/argilosa + PLINTOSSOLO HÁPLICO Distrófico típico, textura argilosa
	Carumbiara	PLINTOSSOLO HÁPLICO Distrófico típico, textura argilosa + LATOSSOLO AMARELO Distrófico típico, textura média + ARGISSOLO VERMELHO AMARELO Distrófico típico, textura média/argilosa + + LATOSSOLO VERMELHO AMARELO Distrófico típico, textura argilosa
Pará	Santarém	LATOSSOLO AMARELO Distrófico típico, textura muito argilosa + NEOSSOLO QUARTZARÊNICO Órtico.
	Itaituba	LATOSSOLO AMARELO Distrófico típico, textura argilosa
	Cachoeira Porteira	ARGISSOLO VERMELHO AMARELO Distrófico típico, textura média/argilosa + ARGISSOLO AMARELO Distrófico plântico, textura média/argilosa + LATOSSOLO VERMELHO AMARELO Distrófico típico, textura argilosa + CAMBISSOLO HÁPLICO Tb Distrófico, textura argilosa.
	Oriximiná	LATOSSOLO AMARELO Distrófico típico, textura média + NEOSSOLO QUARTZARÊNICO Órtico
	Monte Alegre	ARGISSOLO VERMELHO AMARELO Distrófico típico, textura média/argilosa + ARGISSOLO VERMELHO AMARELO Eutrófico típico, textura média/argilosa.
	Altamira	LATOSSOLO AMARELO Distrófico típico, textura muito argilosa + NITOSSOLO VERMELHO Eutrófico típico, textura muito argilosa + ARGISSOLO VERMELHO AMARELO Distrófico típico, textura média/argilosa
	Caxiuanã	LATOSSOLO AMARELO Distrófico típico, textura argilosa + ARGISSOLO AMARELO Distrófico típico, textura média/argilosa + LATOSSOLO AMARELO Distrófico típico, textura média + PLINTOSSOLO HÁPLICO Distrófico típico, textura média/argilosa
	São Félix do Xingú	ARGISSOLO VERMELHO Distrófico típico, textura média/argilosa + ARGISSOLO VERMELHO Eutrófico típico, textura média/argilosa + NITOSSOLO VERMELHO Eutrófico típico, textura muito argilosa + LATOSSOLO VERMELHO AMARELO Distrófico típico, textura muito argilosa
	Carajás	LATOSSOLO VERMELHO AMARELO Distrófico típico, textura média + LATOSSOLO VERMELHO AMARELO Distrófico concrecionário, textura argilosa + ARGISSOLO VERMELHO AMARELO Distrófico típico, textura média/argilosa
	Baião	ARGISSOLO VERMELHO AMARELO Distrófico típico, textura média/argilosa + LATOSSOLO VERMELHO AMARELO Distrófico típico, textura argilosa + ARGISSOLO VERMELHO AMARELO Distrófico típico, textura média/argilosa
	Marabá	ARGISSOLO VERMELHO AMARELO Distrófico cascalhento, textura média/argilosa + LATOSSOLO VERMELHO AMARELO Distrófico típico, textura muito argilosa + CAMBISSOLO CRÔMICO Distrófico, cascalhento, textura argilosa.
	Marajó	NEOSSOLO QUARTZARÊNICO Órtico típico + PLINTOSSOLO HÁPLICO Distrófico típico, textura arenosa/média
	Quatipuru	LATOSSOLO AMARELO Distrófico típico, textura média + ARGISSOLO AMARELO Distrófico típico, textura arenosa/média

	Belém	LATOSSOLO AMARELO Distrófico típico, textura média + + ARGISSOLO AMARELO Distrófico típico, textura arenosa/média
	Barcarena	LATOSSOLO AMARELO Distrófico típico, textura média + ARGISSOLO AMARELO Distrófico típico, textura arenosa/média + ARGISSOLO AMARELO Distrófico típico, textura média/argilosa
Roraima	Boa Vista	LATOSSOLO AMARELO Distrófico típico, textura média + LATOSSOLO AMARELO Distrófico concrecionário, textura argilosa
	Rio Branco	LATOSSOLO AMARELO Distrófico típico, textura argilosa + LATOSSOLO AMARELO Distrófico típico, textura média
Amapá	Macapá	LATOSSOLO AMARELO Distrófico típico, textura argilosa + LATOSSOLO AMARELO Distrófico concrecionário, textura argilosa.

Table 2. Physical and chemical characteristics of the soils in the ADE archaeological sites in Brazilian Amazonia.

Horiz.	Prof. (cm)	Colors	g kg ¹ of soil				pH		cmolc kg ¹ of soil							%	g kg ¹ of soil		mg kg ¹ of soil
			Total sand	Silt	Total clay	Silt/ Clay	H ₂ O	? pH	Ca ⁺⁺	Mg ⁺⁺	K ⁺	S	Al ⁺⁺	CTC E	CTC 1	V	C	N	P
CAMBISSOLO CROMICO Tb Eutrófico antropogênico, textura argilosa. Perfil 1. (Kern 1988) Cachoeira Porteira, Pará.																			
AI	0-7	N2/	550	150	300	0.50	4.6	- 0.6	16.20	6.70	0.12	23.00	1.10	24.10	38.40	60	77.0	4.20	223
A2	7-27	N2/	560	130	310	0.40	5.1	- 1.0	11.40	1.20	0.01	12.60	1.00	13.60	26.30	48	61.0	2.70	321
A3	28-44	N2/	600	130	270	0.50	5.2	- 1.1	9.40	0.40	0.01	9.80	0.90	10.70	17.90	55	37.0	1.50	277
A4	44-53	10YR3/1	540	190	270	0.70	5.2	- 1.1	7.10	0.20	0.01	7.30	0.70	8.00	13.60	54	27.0	0.70	139
AB	53-110	10YR4/2	530	100	370	0.30	4.6	- 0.6	7.30	0.80	0.01	8.00	0.60	8.60	11.60	69	13.0	0.70	86
Bi1	110-148	10YR6/6	420	100	480	0.20	5.0	- 1.0	6.60	0.60	0.01	7.30	0.70	8.00	10.20	72	5.0	0.30	74
Bi2	148-180+	10YR6/6	420	100	480	0.20	5.1	- 1.0	6.60	0.40	0.01	7.30	0.40	7.70	9.70	75	5.0	0.50	80
ARGISSOLO AMARELO Distrófico antrópico, textura argilosa/muito argilosa. Perfil 3. (Kern 1996) Caxiuanã, Pará.																			
Ap	0-4	10YR2/1	470	250	280	0.89	5.8	-	9.30	2.80	0.60	13.00	0	13.00	16.00	81	40.90	-	1001
A2	4-10	10YR2/1	450	220	330	1.50	5.2	-	4.80	1.40	0.30	6.50	1.10	6.60	15.50	42	30.60	-	1175
A3	10-20	10YR2/1	370	170	460	0.37	5.0	-	1.70	0.60	0.14	2.40	1.00	3.40	13.40	18	23.10	-	479
BA	20-30	10YR2,5/2	270	110	620	0.17	4.6	-	0.30	0.30	-	0.60	1.70	2.30	8.40	7	15.00	-	572
Bt1	30-45	10YR4/4	210	100	690	0.14	4.6	-	0.50	0.20	0.01	0.80	1.80	2.60	7.70	10	9.10	-	433
Bt2	45-73	10YR4/6	150	70	780	0.09	4.7	-	-	0.20	-	0.20	1.80	2.00	4.70	4	6.10	-	351
Bt3	73-93	7,5YR5/6	160	100	740	0.13	4.6	-	-	0.10	-	0.10	1.70	1.80	7.60	1	3.60	-	354
Bt4	93-100	7,5YR4/6	170	110	720	0.15	5.0	-	0.50	0.60	-	1.10	1.10	2.20	1.10	27	6.20	-	350
ARGISSOLO AMARELO Distrófico antropogênico, textura média/argilosa. Perfil 3. (Kern 1996) Caxiuanã, Pará.																			
A1	0-7	N2/	610	260	130	2.00	4.8	-	10.20	2.60	0.66	13.50	0	13.50	20.70	65	40.40	-	1250
A2	7-21	N2/	540	260	200	1.30	5.0	-	5.60	1.20	0.12	6.90	0.20	7.10	19.80	35	35.20	-	559
AB	21-42	7,5YR2/1	580	230	210	1.09	5.2	-	2.20	0.50	0.01	2.70	1.10	3.80	12.30	22	10.20	-	453
BA	42-66	7,5YR2/1	410	210	380	0.43	5.0	-	1.20	0.50	-	1.70	2.30	4.00	8.30	20	7.40	-	137
Bt	66-147	7,5YR5/6	230	210	560	0.37	5.1	-	1.20	0.40	-	1.60	2.20	3.80	6.70	24	3.30	-	152
NEOSSOLO LITÓLICO Ta Eutrófico típico, textura média. Perfil 7. (Falesi, 1970) Monte Alegre, Pará.																			
A1	0-30	10YR 2/1	540	280	180	1.55	6.2	- 0.8	37.50	6.14	0.40	44.18	0.10	44.28	49.51	89	46.30	3.80	324
Ae	30-45	10YR 2/2	520	300	180	1.67	5.8	- 0.4	39.60	4.22	0.15	44.13	0.10	44.23	50.26	88	26.90	2.00	471
CAMBISSOLO Ta Eutrófico antropogênico endoconcrecionário, textura média. Perfil 4. (Kern, 1996) Caxiuanã, Pará.																			
A1	0-9	N2/	630	300	70	4.28	5.0	-	10.70	1.40	0.24	12.30	0	12.30	18.00	68	29.40	-	835
A2	9-18	N2/	560	360	80	4.50	5.5	-	14.50	0.50	0.11	15.10	0	15.10	20.80	73	-	-	827
A3	18-25	N2/	600	280	120	2.33	5.9	-	13.10	0.20	0.09	13.40	0	13.40	17.90	75	1.64	-	882
AB	25-42	10YR2/1	490	410	100	4.10	6.2	-	9.20	0.10	0.07	9.40	0	9.40	13.00	72	-	-	110
Bi1	42-60	10YR3/2	410	410	180	2.27	6.4	-	7.00	0.10	0.04	7.00	0	7.00	8.80	80	-	-	243

Bi2	60-88	10YR4/4	310	480	210	2.28	6.3	-	7.10	0.10	0.03	7.80	0	7.80	10.10	77	2.30	-	88
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P = Determinated by Mehlich method (Embrapa, 1997) SB = Sum of cations; V% = base saturation percent

Table 2. Physic and chemical characteristics of the soils in the archaeological sites of ADE in Brazilian Amazonia.

Horiz.	Prof. (cm)	Colors	g kg ¹ of soil				pH		cmolc kg ¹ of soil							%	g kg ¹ of soil		mg kg ¹ of soil
			Total sand	Silt	Total clay	Silt/ Clay	H ₂ O	? pH	Ca ⁺⁺	Mg ⁺⁺	K ⁺	S	Al ⁺⁺	CTC E	CTC 1	V	C	N	P

ARGISSOLO AMARELO Eutrófico antropogênico, textura média /argilosa. Perfil 5. (Kern, 1996) Caxiuana, Pará.

A1	0-6	N2	420	370	210	1.76	5.3	-	15.60	2.60	0.55	18.80	0	18.80	20.80	90	29.70	-	2201
A2	6-28	N2	360	380	260	1.46	5.7	-	9.70	1.90	0.25	11.90	0	11.90	15.50	77	20.00	-	2090
AB	28-38	10YR2/1	390	340	270	1.26	6.0	-	8.00	1.40	0.22	9.70	0	9.70	13.20	74	13.50	-	1715
BA	38-65	10YR2/1	220	280	500	0.56	6.3	-	3.80	0.70	0.05	4.50	0	4.50	7.00	64	7.50	-	1100
Bt	65-147	7.5YR4/6	220	370	410	0.90	6.2	-	3.30	0.50	0.03	3.80	0	3.80	5.90	64	2.30	-	485

LATOSSOLO AMARELO Eutrófico antropogênico, textura argilosa. Perfil 10. (Oliveira Jr., 2002a) Trairão, Pará.

A1	0-37	N2/	598	218	184	1.18	6.7	-0.5	14.30	1.40	0.45	16.40	0	16.40	19.50	84	243.00	2.00	7455
A2	37-62	10YR2/1	631	165	204	0.81	6.7	-0.9	9.70	0.70	0.26	10.80	0	10.80	12.90	84	142.00	0.70	5309
A3	62-95	10YR3/1	616	141	243	0.58	6.8	-1.0	5.50	0.60	0.10	6.30	0	6.30	7.80	81	71.00	0.60	2598
AB	95-129	10YR3/2	550	146	304	0.48	6.9	-1.0	4.10	0.60	0.07	4.80	0	4.80	5.80	83	50.00	0.40	1016
BA	129-177	10YR4/2	486	129	385	0.33	6.9	-1.1	3.70	0.60	0.06	4.40	0	4.40	5.40	81	33.00	0.40	508
Bw1	177-217	10YR5/4	450	125	425	0.29	6.5	-1.1	2.40	0.40	0.07	2.90	0	2.90	3.90	74	14.00	0.30	260
Bw2	217-257	10YR5/8	468	107	425	0.25	6.5	-1.1	2.20	0.40	0.05	2.50	0	2.50	3.30	76	13.00	0.30	294

LATOSSOLO AMARELO Distrófico antropogênico, textura argilosa. Perfil 13. (Oliveira Jr., 2002b) Itaituba, Pará.

A1	0-20	10YR2/2	459	131	410	0.32	6.4	-1.0	9.70	1.00	0.38	11.10	0	11.10	15.00	74	172.0	1.70	158
A2	20-34	10YR3/2	431	121	448	0.27	6.0	-1.3	5.40	0.80	0.18	6.40	0	6.40	12.60	51	112.0	0.90	169
AB	34-51	7.5YR3/4	432	101	467	0.22	5.6	-1.1	2.70	0.60	0.09	3.40	0	3.40	8.70	39	60.0	0.60	203
BA	51-74	7.5YR4/6	414	79	507	0.16	5.5	-1.1	1.80	0.50	0.05	2.40	0	2.40	7.20	33	46.0	0.50	220
Bw1	74-111	7.5YR4/6	374	77	549	0.14	5.4	-0.9	1.90	0.50	0.05	2.50	0	2.50	6.80	37	38.0	0.50	265
Bw2	111-143	5YR4/6	356	94	550	0.17	5.5	-0.9	1.90	0.40	0.04	2.40	0	2.40	6.10	39	33.0	0.30	197
Bw3	143-180	5YR4/6	366	85	549	0.15	5.4	-0.7	1.80	0.40	0.04	2.30	0	2.30	5.70	40	31.0	0.30	271

NEOSSOLO QUARTZARENICO órtico antropogênico. (IDESP, 1974). Marajó, Pará.

A1	0-13	10YR3/2	830	120	50	2.40	5.9	-1.2	2.26	0.73	0.05	3.12	0	3.12	5.44	57	1.40	0.20	929
A2	13-47	10YR3/1	800	140	60	2.33	5.3	-1.0	4.50	0.82	0.17	5.88	0.20	5.88	13.09	43	4.60	0.60	3110
A3	47-85	10YR3/1	800	140	60	2.33	5.2	-0.8	4.09	0.98	0.38	5.84	0.31	5.84	12.79	43	12.90	1.30	2232
AC	85-110	10YR4/3	840	110	50	2.20	5.9	-1.4	3.08	0.49	0.06	3.73	0	3.73	7.41	50	2.30	0.20	1540
C	110-170	2.5YR4/8	820	120	60	2.00	5.7	-1.7	4.57	0.49	0.07	5.31	0	5.31	11.03	48	5.30	0.50	3207

ARGISSOLO AMARELO Distrófico antropogênico, textura média/muito argilosa. Perfil 7. (Rodrigues et al., 1991) Iranduba, Amazonas.

A1	0-18	10YR3/1	570	150	280	0.54	5.1	-0.5	6.30	0.4	0.10	6.90	0	6.90	12.40	56	10.20	1.10	240
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A2	18-44	10YR3/2	580	80	340	0.24	5.6	-0.9	4.90	0.4	0.02	5.40	0	5.40	9.90	55	10.40	1.10	504
AB	44-71	10YR3/3	560	80	360	0.22	5.7	-1.0	3.70	0.2	0.01	3.90	0	3.90	7.40	53	6.90	0.80	132
BA	71-106	10YR5/4	420	70	510	0.14	5.5	-0.8	2.50	0.6	0.01	3.10	0	3.10	6.00	52	6.10	0.70	120
Bt1	106-140	10YR5.5/8	310	50	640	0.08	5.2	-0.7	2.00	0.2	0.01	2.20	0	2.20	5.20	42	3.30	0.60	96
Bt2	140-210	10YR5/8	280	70	650	0.11	5.2	-0.7	1.60	0.2	0.01	1.80	0	1.80	4.60	39	2.00	0.50	168
Bt3	210-270	7.5YR5/8	270	60	670	0.09	5.0	-0.4	1.60	0.2	0.01	1.80	0	1.80	4.60	39	1.80	0.40	204

P = Determinated by Mehlich method (Embrapa, 1997) SB = Sum of cations; V% = base saturation percent

Table 2. Physic and chemical characteristics of the soils in the archaeocological sites of ADE in Brazilian Amazonia.

Horiz.	Prof. (cm)	Cores	g kg ¹ of soil				pH		cmolc kg ¹ of soil							%	g kg ¹ of soil		mg kg ¹ of soil
			Total sand	Silt	Total clay	Silt/Clay	H ₂ O	? pH	Ca ⁺⁺	Mg ⁺⁺	K ⁺	S	Al ⁺⁺	CTC E	CTC 1	V	C	N	P

LATOSSOLO AMARELO Distrófico antropogênico, textura muito argilosa. Perfil 6. (Silva et al., 1970) Manacapuru, Amazonas.

A1	0-8	10YR3/1	480	150	370	0.40	5.5	-0.6	9.39	1.91	0.58	11.40	0.21	11.61	18.52	62	23.30	1.80	240
A2	8-31	10YR3/1	440	130	430	0.30	5.7	-1.0	8.80	0.67	0.19	9.56	0.21	9.77	19.31	49	21.70	1.20	155
AB	31-63	10YR3/2	380	120	500	0.24	5.7	-1.1	5.96	0.58	0.04	6.60	0.21	6.81	14.37	46	13.30	0.70	246
Bw1	63-123	10YR6/6	310	40	650	0.06	5.5	-0.8	1.60	0.65	0.03	2.30	0.20	2.50	5.06	46	3.10	0.40	116
Bw2	123-160	10YR6/8	270	50	680	0.07	5.3	-0.6	1.56	0.41	0.02	2.03	0	2.03	5.01	41	1.80	0.40	215

ARGISSOLO AMARELO Eutrófico antropogênico, textura média/muito argilosa Perfil 1 (Santarém)

AP	0-20	7,5 YR 2/1	290	430	280	1.53	6.1	-0.5	47.91	7.09	0.42	55.56	0	55.56	64.17	87	36.40	4.90	197
A ₂	20-45	7,5 YR 3/2	290	350	360	0.97	6.6	-0.8	48.48	4.76	0.15	53.50	0	53.50	59.77	90	26.70	3.40	219
BA	45-110	7,5 YR 4/2	130	180	690	0.26	6.4	-0.9	23.98	2.13	0.07	26.24	0	26.24	30.61	86	10.60	1.10	154
Bt	110-150	7,5 YR 5/8	100	130	770	0.17	6.1	-0.6	8.58	1.82	0.06	10.50	0	10.50	13.41	78	8.20	0.90	47

CAMBISSOLO HÁPLICO Tb Eutrófico antropogênico, textura argilosa Perfil 16 (Borda do Planalto – CBA) Santarém

A ₁	0-16	5 YR 3/1	220	450	330	1.36	6.4	-0.9	37.27	16.46	0.15	54.03	0.11	54.14	61.82	87	55.60	4.60	77
AB	16-34	5 YR 3/2	130	480	490	0.98	5.9	-0.6	9.21	0.92	0.03	10.22	0.21	10.43	14.38	71	12.00	0.90	90
BA	34-61	5 YR 4/2	180	400	420	0.95	6.0	-0.9	5.33	0.92	0.02	6.32	0.21	6.53	9.60	66	7.30	0.60	65
Bi ₁	61-95	5 YR 4/6	240	390	370	1.05	6.0	-0.7	3.17	0.61	0.02	3.86	0.20	4.06	6.26	62	3.90	0.40	68
Bi ₂	95-143	5 YR 5/8	180	370	450	0.82	5.8	-0.4	2.10	0.61	0.03	2.80	0.20	3.00	5.02	56	2.80	0.40	76

LATOSSOLO VERMELHO AMARELO Distrófico antropogênico, textura muito argilosa Perfil 17 (Andicobal) Santarém, Pará

A ₁	0-13	5 YR 2/1	140	430	430	1.00	7.1	-0.8	31.80	6.27	0.11	38.26	0.11	38.36	40.67	94	48.10	4.10	26
A ₂	13-29	5 YR 2/1	200	320	490	0.65	6.5	-0.8	23.56	2.94	0.06	26.63	0.21	26.84	31.93	83	33.80	2.40	30
AB	29-50	5 YR 3/2	100	320	580	0.55	6.1	-0.9	13.52	1.87	0.03	15.48	0.21	15.49	21.42	72	19.70	1.40	43
Bw ₁	50-73	5 YR 4/2	70	140	790	0.18	5.3	-0.9	5.04	0.61	0.03	5.72	0.10	5.82	15.41	37	11.20	0.90	33

Bw ₂	73-106	5 YR 4/4	90	280	630	0.44	5.2	-0.9	2.90	0.40	0.02	3.37	0.31	3.68	7.77	43	7.40	0.70	24
Bw ₃	106-145	5 YR 4/8	160	310	530	0.58	5.1	-0.5	2.15	0.61	0.03	2.85	0.42	3.27	6.64	43	5.80	0.60	47

LATOSSOLO AMARELO Eutrófico antropogênico, textura muito argilosa. Perfil 3EI. Rodrigues et al. (2000) Apuí, Amazonas.

Ap	0-18	10YR2/1	253	248	499	0.50	6.5	-1.1	13.50	1.30	0.58	15.40	0	15.40	24.10	64	33.60	3.00	260
A2	18-34	10YR3/1	254	204	542	0.38	6.2	-1.2	9.10	0.80	0.19	10.10	0	10.10	17.50	58	18.10	1.50	300
AB	34-57	10YR3/2	175	145	680	0.21	6.0	-1.1	5.00	0.60	0.04	5.60	0	5.60	10.00	56	7.80	0.90	160
BA	57-94	10YR5/6	141	104	755	0.14	6.0	-1.1	3.30	0.80	0.03	4.10	0	4.10	7.50	55	4.00	0.70	120
Bw1	94-145	10YR5/8	145	100	755	0.13	6.0	-1.1	2.70	0.90	0.02	3.60	0	3.60	7.20	60	2.70	0.60	160
Bw2	145-197	10YR6/6	144	103	753	0.14	5.9	-1.0	2.20	0.60	0.02	2.80	0	2.80	6.40	44	2.10	0.40	140
Bw3	197-280	10YR6/6	125	122	753	0.16	6.0	-0.7	1.70	0.70	0.02	2.40	0	2.40	4.80	50	2.10	0.50	42

P = Determinated by Mehlich method (Embrapa, 1997) SB = Sum of cations; V% = base saturation percent